

"We should stop training radiologists now. Its just completely obvious that within five years, deep learning is going to do better than radiologists."

– Geoffrey Hinton (in 2016)

1 Introduction

Artificial intelligence (AI) is a general-purpose technology with transformative potential similar to that of the steam engine and electricity (Brynjolfsson and Mitchell, 2017; Brynjolfsson et al., 2017; Agrawal et al., 2018; Acemoglu and Johnson, 2023; Goldfarb et al., 2023; Frank et al., 2019). But, in contrast to the transformation during the industrial revolutions, AI can potentially displace humans from tasks that require complex reasoning (Webb, 2019; Felten et al., 2019; Brynjolfsson and Mitchell, 2017). Advances in AI have therefore caused concern about the role of human work, even in highly skilled occupations (Ford, 2015). Indeed, a growing literature shows that AI tools based on machine learning can outperform humans on a host of predictive tasks, including those typically performed by experts (Liu et al. 2019; Lai et al. 2021; Mullainathan and Obermeyer 2019; Kleinberg et al. 2017).

However, Hinton’s prediction about radiology — an iconic example in machine learning — may have been premature.¹ Many argue that instead of being replaced by AI, it is optimal to have human radiologists use AI assistance, at least for the foreseeable future (Langlotz, 2019; Agrawal et al., 2019). In addition to considerable legal and regulatory challenges that stand in the way of full automation, there may be potential gains from combining human expertise with AI input that cannot be realized by exclusively relying on one or the other. For example, radiologists may correct mistaken AI predictions or may have access to information about the clinical context that has not yet been systematized and built into AI tools. Current regulatory practice is consistent with these arguments – approved AI tools for clinical decision-making typically play a supporting role rather than operating autonomously (see Norden and Shah, 2022; Harvey and Gowda, 2020, for example). Similar reasons are likely to result in partial task automation or collaboration between human experts and AI being optimal in other contexts.

We run an experiment to examine how radiologists use AI predictions, whether AI assistance improves their performance, estimate a model of (potentially biased) belief updating, and analyze what this model implies about the optimal form of collaboration between AI and humans.² If humans correctly combine AI predictions with their own information, then AI

¹A more nuanced but qualitatively similar prediction that machine learning tools will displace radiologists is conveyed in Obermeyer and Emanuel (2016).

²We will use the terms ‘humans’, ‘radiologists’, and ‘participants’ interchangeably.

only list the patient’s primary symptom (see appendix B.2 for examples). Irrespective of the pathology suspected by the treating physicians, radiologists are expected to report all pathological findings.

Because image-based classification is a core task performed by radiologists, a high-paying profession, it is not surprising that AI tools have made significant inroads in the field in the last decade. Recent advances in deep learning methods for image recognition have yielded algorithms that can match or surpass the performance of human radiologists (Obermeyer and Emanuel, 2016; Langlotz, 2019). As of 2020, 55 companies offered a total of 119 algorithmic products of which 46 have FDA approval (Tadavarthi et al., 2020). Most products related to clinical decision-making are marketed as support tools as opposed to autonomous tools, partly due to regulatory and liability issues (Harvey and Gowda, 2020).

3.1.2 *CheXpert*

We provide AI assistance using the CheXpert model, which is a deep learning prediction algorithm for chest X-ray interpretation (Irvin et al., 2019). This model is trained on a dataset of 224,316 chest radiographs of 65,240 patients labeled for the presence of fourteen common chest radiographic pathologies.⁹ The algorithm does not use any other patient information, such as the clinical history or vitals.¹⁰ Nonetheless, a prior version of this algorithm was shown to match or surpass the performance of board-certified radiologists from Stanford Hospital on five pathologies (Patel et al., 2019). These study results are also presented to our participants when introducing the AI tool. Section 4 confirms that the algorithm outperforms a majority of radiologists in our experiment. We relegate additional details about the algorithm to appendix B.3. The algorithm assistance to our participants will be in the form of a vector of probabilities for the presence of every pathology.¹¹

⁹The term artificial intelligence is typically reserved for a system of different prediction tasks to mimic a more complex set of behaviors, whereas machine learning is concerned with one specific prediction task. For a detailed discussion of this distinction see, for instance, (Taddy, 2018).

¹⁰While large datasets of images are increasingly available (e.g. Kramer et al. 2011, Johnson et al. 2016, Irvin et al. 2019) it is significantly more difficult to construct such datasets for other patient information due to the compulsory manual review of textual data for HIPPA compliance.

¹¹Some algorithms attempt to make their predictions explainable to a human by highlighting the parts of the image that drive a specific prediction. However, prior studies show that providing such localization in addition to the numeric output does not improve the accuracy of radiologists (Gaube et al., 2022). A quantitative output allows us to compute a Bayesian benchmark to the radiologist’s prediction, which is otherwise difficult.

4.3 Discussion

We find that AI assistance does not improve the performance of our participants on average, even though the AI predictions are more accurate than the majority of radiologists, and that radiologists respond to this assistance. However, the average treatment effects mask important heterogeneity – AI assistance improves performance for a large set of cases, but also decreases performance in many instances. These results point to biases in the use of AI predictions which we will further investigate in section 5.

We also find that humans have access to valuable contextual information, suggesting that full automation has its drawbacks. But, the biases above – especially in conjunction with our finding that radiologists take longer when given AI assistance – undercut this potential information advantage in a setup that involves AI assistance. Thus, the problem of how best to deploy AI assistance may be non-trivial and the optimal solution may involve selective automation and/or AI assistance. Section 6 analyzes this problem.

Appendix C.4 shows that the results are qualitatively robust to a variety of alternative analyses. The results are similar when we split the analysis by experimental design and when we only focus on the initial across-comparison (using design 1 and 2), although the latter leads to estimates that are imprecise. Alternative ground-truth measures – including a leave-one-out ground truth based on the assessments of our experimental participants and a continuous ground truth, which simply averages the assessments of the ground-truth labelers – also yield similar conclusions. The qualitative patterns of the treatment effects are unchanged if we calibrate each radiologists’ assessments to the ground-truth before conducting the analysis. Finally, incentives for accuracy, which are cross-randomized in designs 1 and 3, do not have significant effects either.

5 Automation/Own-Information Bias/Neglect

An upshot of the results in section 4 is that our participants deviate from the baseline of a Bayesian with correct beliefs about the joint distribution of their own information and the AI signal. In this section, we model and estimate systematic deviations from this benchmark – which we will refer to as Bayesian²⁵ for short – and determine the implications of these deviations on utilizing human expertise and AI predictions.

²⁵The omission of the qualifier “with correct beliefs” slightly abuses terminology because a possible explanation of the deviations we have documented is that our participants are Bayesians but update their beliefs using an incorrect model for the joint distribution of s^A , s^E and ω . We will entertain this possibility below.

highly-skilled profession that is being transformed by AI, radiology also represents a large class of professionals whose main job is a high-stakes classification task. Since we can simulate radiologists’ normal workflow, this is an ideal setting for conducting such an experiment.

While the potential benefits of deploying AI assistance are large in our setting, biases in humans’ use of AI assistance eliminate these potential gains. Even though the AI tool in our experiment performs better than two thirds of radiologists, optimally combining their predictions could substantially improve performance. Yet, we find that giving radiologists access to AI predictions does not, on average, lead to higher performance. This average treatment effect, however, masks systematic heterogeneity: providing AI does improve radiologists’ predictions and decisions for cases where the AI is certain, but not when uncertain. This latter result – that prediction quality can be reduced for some range of AI signals – rejects Bayesian updating with correct beliefs. We also describe systematic errors in belief updating – namely radiologists exhibit automation neglect (e.g. radiologists weight the AI prediction relative to their own) and treat the AI prediction and their own signals as independent even though they are not. Moreover, radiologists take significantly more time to make a decision when AI information is provided.

Together, these results have important implications for how to design the collaboration between humans and machines. Increased time costs and sub-optimal use of AI information both work against having radiologists make decisions with AI assistance. In fact, an optimal delegation policy that utilizes heterogeneity in treatment effects given the AI prediction suggests that cases should either be decided by the AI alone or by the radiologist alone. Only a small share of cases are optimally delegated to radiologists with access to AI. In other words, we find that radiologists should work *next to* as opposed to *with* AI. To the extent that expert decision makers generally under-respond to information other than their own (Conlon et al., 2022) and incorporating additional information is cognitively costly, these insights may hold in other settings where experts’ main job is a classification task.

There are several important considerations that are outside the scope of this work. One question motivated by the unrealized potential gains of AI assistance we find is whether the benefits from AI-specific training for radiologists and/or experience with AI are large. Answering these questions requires different experimental designs or longer-run studies. Another open question is whether the heterogeneity in the value of AI assistance is correlated with a human’s baseline skill or other characteristics, and whether such correlation can be predicted to target assistance. A consideration in the organization of human-AI collaboration that we abstract away from is that the form of collaboration may influence experts’ incentives, to which humans may respond strategically. Finally, the use of AI in practice will be